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Realizing Ultrahigh-Q Resonances Through Harnessing Symmetry-Protected Bound States in the Continuum

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Harnessing the power of symmetry-protected bound states in the continuum (SP BICs) has become a focal point in scientific exploration, promising many interesting applications in nanophotonics. However, the practical realization of ultrahigh quality (Q) factor quasi-BICs (QBICs) is hindered by the fabrication imperfections. In this work, an easy approach is proposed to achieve ultrahigh-Q resonances by strategically breaking symmetry. By introducing precise perturbations within the zero eigenfield region, QBICs with consistently ultrahigh-Q factors, beyond conventional limitations are achieved. Intriguingly, intentionally disrupting symmetry in the maximum eigenfield region leads to a rapid decline in QBIC's Q-factors as the asymmetry parameter increases. Leveraging this design strategy, ultrahigh-Q modes with a high Q-factor of 36,694 in a silicon photonic crystal slab are experimentally realized . The findings establish a robust and straightforward pathway toward unlocking the full potential of SP BICs, enhancing light-matter interactions across diverse applications.

1. Introduction

Bound states in the continuum (BICs) have triggered extensive interest in the past decade because they possess infinite quality (Q) factors and extraordinary field confinement capability.[1–3] Although BICs originate from quantum mechanics,[4,5] it has been demonstrated as a universal phenomenon in wave physics, including electromagnetic wave,[6–8] acoustic wave,[9–13] elastic wave,[14] etc. Essentially speaking, BICs are particular leaky modes with infinite Q-factors despite embedding in the continuum spectrum, which holds great promise in enhancing light-matter interactions.[3,15–20] Besides, BICs can be linked with a polarization singularity[21] opening the door toward tailoring the polarization property.[22–26]

Typically, BICs can be categorized based on the physical mechanisms behind them, including symmetry-protected (SP) BICs,[17,18,27–34] Friedrich–Wintgen BICs,[9–11,35–41] accidental BICs,[6,7,28,42], and Fabry-Perot BICs.[6,12,43–45] Among these BICs, SP BICs are the most intensively studied ones. They can be easily found as long as the unit cell of periodic structures satisfies the inversion-symmetry and mirror-symmetry. Generally speaking, one has to break the symmetry to induce the transition from BICs to quasi-BICs (QBICs)[46] as only QBICs can be excited by external sources like plane wave incidence. It has been widely recognized that the Q-factors of QBICs are inversely proportional to the squared asymmetry parameters. To date, the measured Q-factors of QBICs induced by the broken symmetry ranged from a few hundred to a few thousand,[17,18,32,47–50] as summarized in **Table 1**. Whether it is possible to push the measured Q-factors of symmetry-broken induced QBICs into a few tens of thousands, or even a few hundreds of thousands, still remains unanswered. Although the upper limit of measured Q-factors based on QBICs has been pushed into almost half a million by harnessing topological BICs through merging multiple BICs into a single one,[51] the structure shall be carefully designed. On the other hand, the structure must be suspended to create a symmetric environment. Introducing the substrate will significantly reduce the Q-factors of QBICs.

SP BICs may provide an alternative solution for realizing ultrahigh-Q modes because they still exist in a metastructure on the substrate. Interestingly, there is one common feature for those reported designs: the structure's symmetry broken always results in a significant change in the eigenfield distribution deviating from the case of a BIC. At first glance, it seems quite straightforward that breaking a unit cell's symmetry inevitably triggers the excitation of a QBIC with a high-Q factor. However, not all of the QBICs based on SP BICs can be effectively excited because there are numerous ways of introducing broken symmetry. In other words, different kinds of symmetry broken provide abundant freedom of tailoring the Q-factor of QBICs. Some of

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Table 1. Experimental high-Q resonances in all-dielectric metasurfaces on substrate.

Structure	Unit cell	Resonance Type	Measured Q	Wavelength [nm]	Reference
Si MS	Ring+Bar	EIT	483	1371	[55]
GaAs MS	L-shape	SP BIC	600	980	[31]
Si MS	Two bars-1	SP BIC	170	1460	[18]
HBN MS	Two bars-2	SP BIC	350	790	[49]
Si MS	Disk with hole-1	SP BIC	128	1345	[17]
Si MS	Disk with hole-2	TD BIC	728	1505	[47]
TiO_2 MS			160	685	
Si MS	Disk with hole-3	TD BIC	4,990	1497	[56]
Si MS	Notched disk	SP BIC	1,946	1430	[32]
U-shape	U-shape	SP BIC	3,571	1548	[50]
Si MS	Elliptical dimer	SP BIC	200	6500	[48]
T-shape	T-shape	BIC	18,511	1588	[57]
Si PCS	**Double holes**	**SP BIC**	**36,964**	**1552.5**	**This work**

MS: metasurface; PCS:photonic crystal slab; HBN: hexagonal boron nitride; EIT: electromagnetically induced transparency; SP BIC: symmetry protected bound state in the continuum; TD BIC: toroidal dipole bound state in the continuum

them may lead to the realization of an ultrahigh-Q QBIC without resorting to the delicate engineering of topological charges to create merged BICs.

In this work, we report an easy approach of obtaining ultrahigh-Q modes by leveraging SP BICs through strategically breaking symmetry (**Figure 1**). We show how to break unit cell's symmetry of a metastructure plays a vital role in governing the Q-factor of different QBICs. The nature of BICs hosted in a photonic crystal slab (PCS) is well preserved when the square hole in a unit cell is rotated because such a symmetry broken does not introduce the effective perturbation on the eigenfield of BICs. Only when the eigenfield distributions of BICs are severely disturbed by introducing the additional broken symmetry at the field maximum (called strong perturbation in Figure 1a), the Q-factor of QBICs will reduce to a relatively small value even at a mild asymmetry parameter (Figure 1c). On the contrary, if

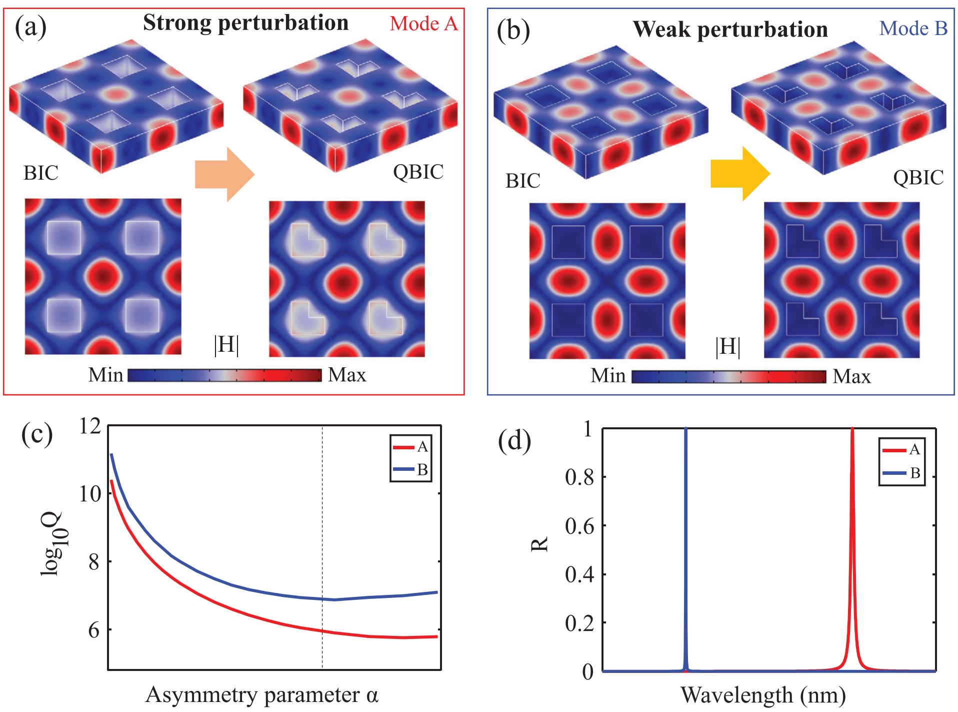

Figure 1. Effect of symmetry broken on Q-factors of QBIC. a) Eigenfield distributions of BIC and QBIC under strong perturbation. b) Eigenfield distributions of BIC and QBIC under weak perturbation. c) Q-factors of QBICs based on modes A and B versus the same asymmetry parameter. d) Typical reflection spectrum of PCS supporting QBICs based on modes A and B.

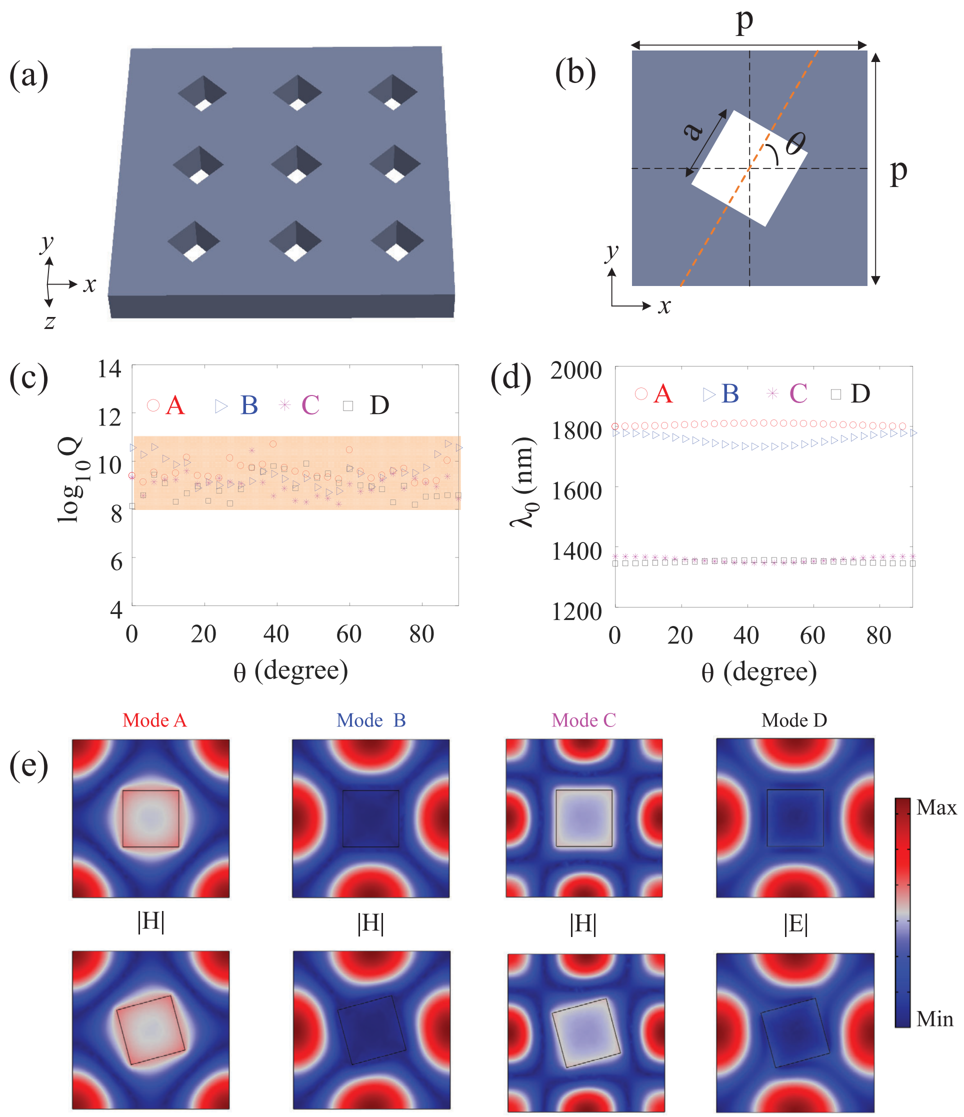

Figure 2. BICs in a PCS with a rotated square hole. a) Schematic of the PCS with rotated square hole. b) Schematic drawing of a unit cell of the PCS in x-o-y plane. c) Q-factors of four modes as a function of rotated angle θ. d) Resonance wavelengths of four modes versus rotated angle θ. e) Eigenfield distributions in the x-o-y plane of four modes for $\theta = 0°$. f) Eigenfield distributions in the x-o-y plane of four modes for $\theta = 15°$. The lattice constants along both *x*- and *y*-axis are p = 700 nm, and the length of the square hole is a = 250 nm. The thickness of silicon thin film is 220 nm.

the structure perturbation is created around zero field region (called weak perturbation in Figure 1b), the Q-factor of QBICs can be maintained at an extremely large value, suggesting a viable way of realizing ultrahigh-Q modes (Figure 1c,d). Thus, by creating a specific perturbation in the weak field region, we can selectively excite a particular QBIC whose Q-factors can maintain above 10^7 even for a large asymmetry parameter. To verify this, we fabricated two sets of 220 nm Si PCSs on a 2 µm-SiO_2/Si substrate carefully designed so that two QBICs can be selectively excited. The measured Q-factors are above 30,000 for less perturbed BICs, while they decrease significantly with the increasing asymmetry parameter for QBICs with a severe perturbation. Our findings may provide a simple but straightforward way of achieving ultrahigh-Q modes and may find exciting applications in boosting light-matter interactions, such as ultralow threshold laser, nonlinear optics, strong coupling, and biosensors.

2. Results and Discussion

We consider a suspended PCS with a square lattice as an example to illustrate how perturbation controls the Q-factors of QBICs. **Figure** 2a,b shows the schematic drawing of a PCS. The unit cell of the PCS is made of a rotated square hole in a continuous silicon film with a thickness of 220 nm. The rotation angle is set as θ. The periodicities along the *x*- and *y*-axis are p = 700 nm. The length of the square hole in the unit cell is a = 250 nm. For $\theta = 0°$, the PCS satisfies the in-plane inverse and mirror-symmetry, thus hosting a series of SP BICs. The top panel of Figure 2e shows the eigenfield distributions of four lowest-order SP BICs, where the

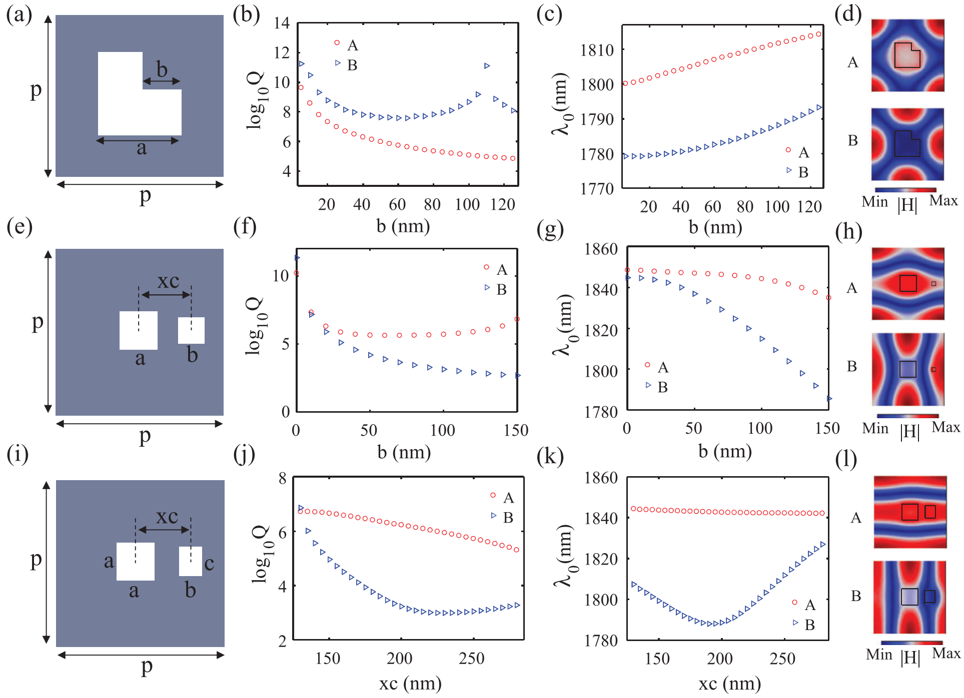

Figure 3. Effect of symmetry broken on Q-factors of QBICs. a) Schematic drawing of a unit cell of PCS with an L-shaped hole. The dimensions of the structure are p = 700 nm and a = 250 nm. b,c) Q-factors b) and resonance wavelengths c) of modes A and B versus parameter b. d) The eigenfield profiles of modes A and B for b = 80 nm. e) Schematic drawing of the unit cell of PCS with double holes. The dimensions of the structure are p = 700 nm, a = 160 nm, and xc = 250 nm. f,g) Q-factors f) and resonance wavelengths g) of modes A and B versus parameter b. h) The eigenfield profiles of modes A and B for b = 40 nm. i) Schematic drawing of a unit cell of PCS with double holes. Only xc is variable, the other parameters are p = 700 nm, a = 160 nm, b = 100 nm and c = 120 nm. j–k) Q-factors j) and resonance wavelengths k) of modes A and B versus parameter b. l) The eigenfield profiles of modes A and B for xc = 200 nm.

former three are TE-like modes, and the latter belongs to a TM-like mode. Surprisingly, when rotating the square hole in an anti-clockwise way, as shown in Figure 2c, the Q-factors of these four BICs are maintained above 10^8 even though the mirror symmetry is no longer satisfied. Meanwhile, the resonance wavelengths show mild change with the rotated angle, as demonstrated in Figure 2d. We also plot the eigenfield distributions of four BICs for $\theta = 15°$ in the bottom panel of Figure 2e. For modes A and C, the magnetic fields inside and around the hole are twisted with the rotated holes while the rest regions are similar to the case of $\theta = 0°$. For modes B and D, since the magnetic fields and electric fields are mainly distributed at the center of four sides, the rotated hole has a negligible effect on the eigenfield profiles. The results shown in Figure 2 suggest that symmetry broken is not the prerequisite of exciting QBICs. A similar phenomenon is also found in a metasurface made of a disk with a rotated square hole (See Figure S1, Supporting Information).

To effectively induce the transition from BICs to QBICs, one has to break the symmetry of eigenfield profiles of BICs. Since different BICs have distinct eigenfield profiles, their Q-factors may show drastically different trends even for the same broken symmetry. To illustrate this notion, we introduce three different symmetry-broken ways to excite modes A and B. The first one is to replace the square hole with an L-shaped hole, as schematically shown in **Figure** 3a. The structure parameters are the same as those in Figure 2. Such a symmetry broken can severely perturb the eigenfield profile of mode A because part of magnetic fields is strongly localized within the square hole, as seen in Figure 3d. Thus, it is expected that Q-factor of mode A reduces significantly with the width b, as demonstrated in Figure 3b. In contrast, the Q-factor decreases much more slowly for mode B and is maintained above 10^8 because its eigenfield profile is weakly perturbed by the hole shape. Due to the reduced area of the hole, the resonant wavelengths of both modes increase slowly with width b, as seen in Figure 3c.

To effectively excite QBIC based on mode B, we need to perturb the eigenfield profile of mode B intentionally. Except for a square hole in the center, another square hole close to the right side of the lattice is created, as schematically shown in Figure 3e, so that the eigenfield around the right side can be significantly disturbed. The width of the square hole in the center is a = 160 nm. For this type of perturbation, the right hole's center is fixed as xc = 250 nm and yc = 0 nm while the width is gradually increased from 0 to 150 nm. Indeed, the Q-factor of mode B drops much faster than that of mode A, as verified in Figure 3f. Besides, the resonant wavelength of mode B shows considerable blueshift with the width, while it decreases mildly for mode A (Figure 3g). The remarkable difference of Q-factors for the two modes is also reflected from the eigenfield profile shown in Figure 3h.

Another way of exciting QBIC based on mode B is to move the right hole along the x-axis (Figure 3i). The right rectangular hole has dimensions of b = 100 nm and c = 120 nm. Note that the eigenfields of modes A and B change after the right hole is introduced. When it is gradually moved into the eigenfield region,

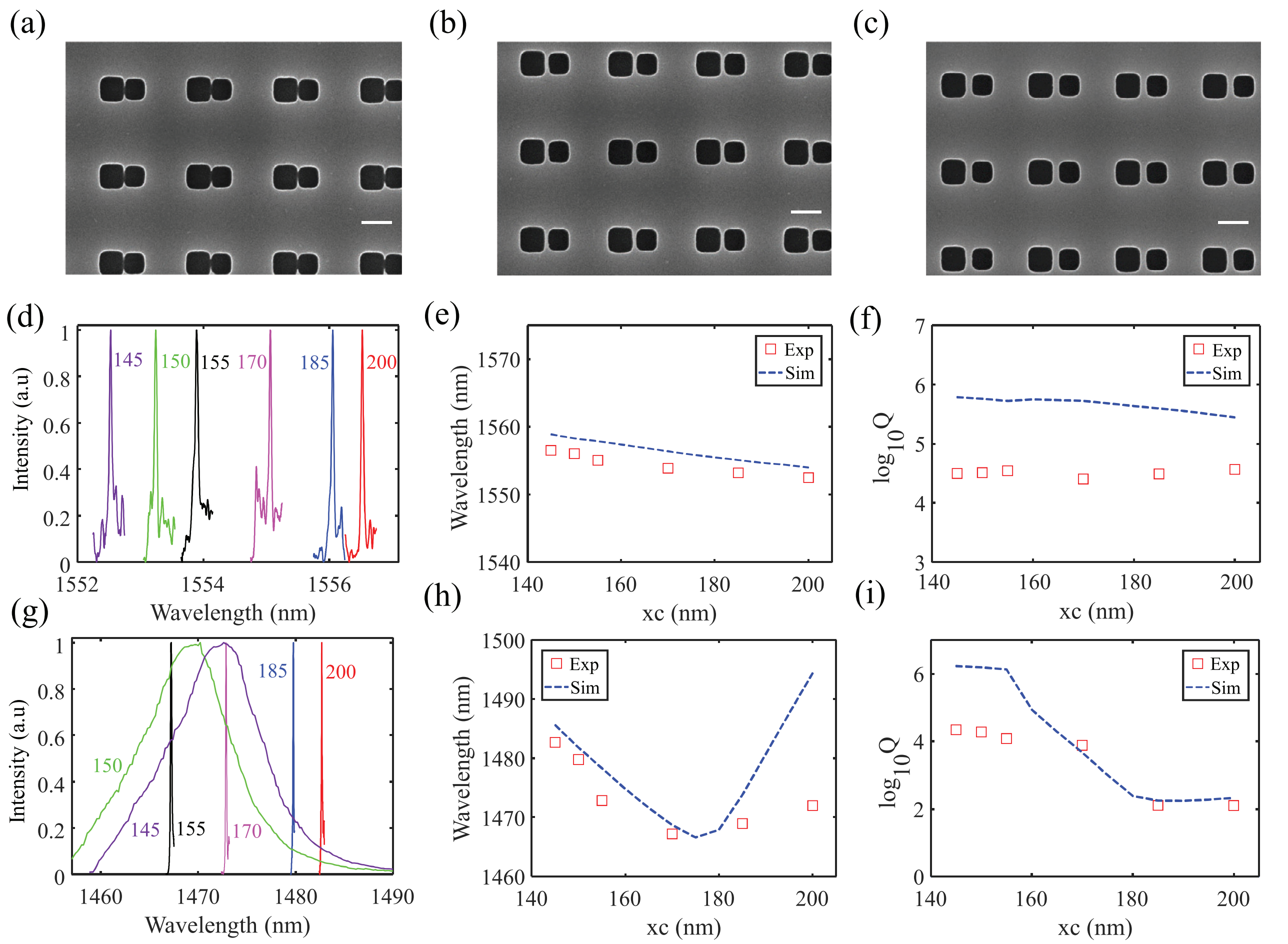

Figure 4. Experimental demonstration of high-Q QBICs based on modes A and B. a–c) SEM images of fabricated PCS with double holes. d) Scattering spectra of PCS while QBIC based on mode A is excited in this wavelength window. e,f) Retrieved resonance wavelengths e) and Q-factors f) of QBICs at different xc. g) Scattering spectra of PCS with double holes while QBIC based on mode B is excited in this wavelength window. h,i) Retrieved resonance wavelengths h) and Q-factors i) of QBIC based on mode B at different xc.

the Q-factor of mode B experiences a considerable reduction with the center xc (Figure 3j). Nevertheless, the Q-factor of mode A changes much more slowly and is maintained above 10^5. Careful examination of eigenfields of both modes at different xc indicates that mode A has almost unchanged eigenfields no matter how xc varies (See Figure S2, Supporting Information). On the contrary, the eigenfields experience significant changes when xc is varied from 130 to 280 nm. This is why the Q-factors of modes A and B show different trends on xc. Here, it is worth noting that the resonant wavelength of mode B does not decrease monotonically with the increasing xc (Figure 3k). It is first reduced to 1788 nm at xc = 190 nm, and then increases with the enlarged xc. This can be attributed to the coupling between two holes. Unlike mode B, mode A has a relatively stable resonance wavelength because the right hole does not influence its eigenfield profile (Figure 3l). Therefore, it can be safely concluded from Figure 3 that the way of introducing symmetry broken has a huge impact on the Q-factors of QBICs. It suggests a good alternative way of exciting an ultrahigh-Q resonance based on an SP BIC: introducing the structure perturbation within the zero field region. Note that the strategy of obtaining high-Q resonances based on QBICs is universal and robust. It can be generalized to the metasurface made of an array of dielectric meta-atoms (i.e., disk) (See Figure S3, Supporting Information).

3. Experimental Demonstration

After gaining a solid understanding of the effect of symmetry-broken on the Q-factor of QBICs, we move to the experimental demonstration of ultrahigh-Q resonances based on SP BICs by creating specific perturbations. We fabricate a series of Si PCSs with double holes and an L-shaped hole based on silicon on an insulator substrate (220 nm-Si/2 μm-SiO_2/Si). The pattern is defined by electron beam lithography followed by inductively coupled plasma etching (See Figure S4, Supporting Information). The lattice constants of structures with double holes are p = 570 nm. The widths of center and right square holes are 165 and 145 nm, respectively. Besides, only the right square hole is moved along the *x*-axis, while the other square hole is fixed in the center. For the structure with an L-shaped hole, the left and bottom sides have the same dimensions a = 300 nm.

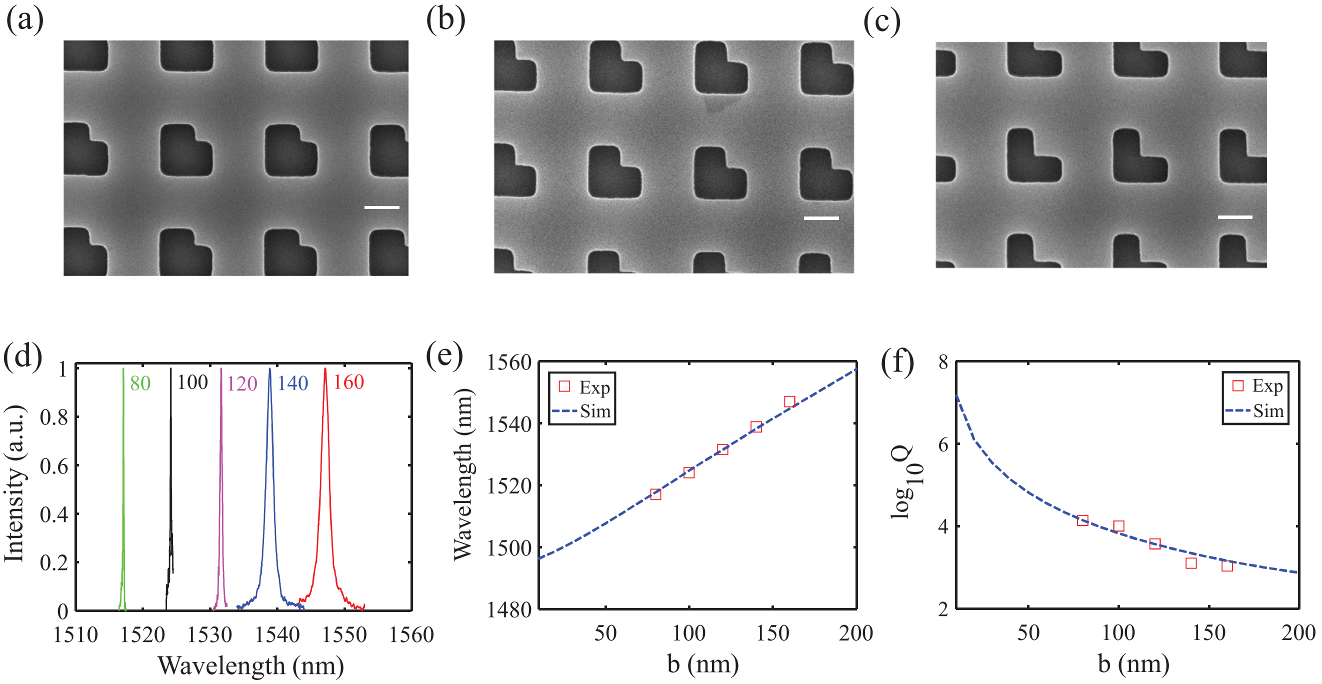

Figure 5. Experimental demonstration of high-Q QBICs in a PCS with an L-shaped hole. a–c). d) Scattering spectra of PCS while QBIC based on mode A is excited in this wavelength window. e,f) Retrieved resonance wavelengths e) and Q-factors f) of QBICs at different widths b.

Figure 4a–c shows the scanning electron microscopy (SEM) image of PCSs with double holes. We apply the home-built microscope spectroscopy system based on cross-polarization measurement (Figure S5, Supporting Information) to characterize such high-Q resonances.[51–53] Figure 4d shows the measured scattering spectra of PCSs with double holes. It is clearly seen that a sharp Fano resonance is displayed for different xc ranging from 145 to 200 nm, corresponding to the excitation of a QBIC based on mode A. The Q-factors and resonance wavelengths of the QBIC, as shown in Figure 4e,f, are retrieved by following the standard Fano-fitting procedure.[54] Remarkably, the measured Q-factors of QBICs based on mode A are maintained above 30,000 at different xc while the resonance wavelengths are quite stable ≈1555 nm. The maximum Q-factor obtained here is up to 36,964. To make a comparison, we also calculated the Q-factors and resonance wavelengths of eigemode A when xc is varied from 145 to 200 nm. Because such a perturbation has a weak effect on the eigenfield profile of mode A, the calculated Q-factors are maintained above 2×10^5 for xc ranging between 145 and 200 nm even when there is substrate underneath the PCS. Meanwhile, the resonance wavelengths remain nearly constant. Good agreements on both Q-factors and resonance wavelengths can be found between simulation and measurement, as demonstrated in Figure 4e,f. Due to the fabrication imperfection, the measured Q-factors show almost one order of magnitude reduction compared to the simulation. Further optimization of fabrication process may push the Q-factor to 10^5 and even more.

Figure 4g shows the scattering spectra of PCSs with double holes dominated by mode B. Obviously, the resonance linewidths broaden with the increased xc, signifying a reduced Q-factor. Following the same fitting procedure, we retrieved the resonance wavelengths and Q-factors shown in Figure 4h,i. The simulation results are also presented to make a fair comparison. Reasonably good agreement between experiments and simulations can be found for both resonant wavelengths and Q-factors. Unlike mode A showing stably high Q-factor at different xc, the Q-factors of the QBIC decrease from 22,533 to 127 as the xc is increased from 145 to 200 nm. Although the maximum measured Q-factor of QBIC based on mode B shows almost two orders of magnitude reduction compared to the calculated one, it is still larger than the maximum Q-factor of symmetry-broken induced QBICs reported experimentally so far. This suggests that moving the hole position is a good way of breaking structure symmetry to engineer the Q-factors.

Next, we consider the PCSs with L-shaped holes. Three typical SEM images are shown in **Figure** 5a–c. The relevant scattering spectra are shown in Figure 5d. It can be found that the resonance wavelengths of mode A show redshifts with the increased width b. Simultaneously, the resonance linewidths broaden as b increases from 50 to 190 nm. The largest Q-factor retrieved from Figure 5d is up to 13,790. Excellent agreements can be found between theory and experiments for both resonance wavelength and Q-factors (Figure 5e,f). For mode B, we did not observe high-Q Fano resonances in the scattering spectra due to the ultrahigh-Q factors of mode B. Simulation results show that the Q-factors of mode B are above half a million (Figure S6, Supporting Information).

Finally, we list the experimental high-Q resonances based on all-dielectric metasurfaces on the substrate in Table 1. Most of the measured Q-factors range from a few hundreds to a few thousands. Our results represent the largest Q-factor based on SP BICs. We believe that the upper limit of the measured Q-factor can be further pushed into 10^5 by choosing the proper perturbation way and optimizing the fabrication process.

4. Conclusion

In summary, we propose a simple approach for constructing ultrahigh-Q resonances based on SP BICs without resorting to topological charge engineering of BICs. Notably, we show that the way of symmetry breaking plays a crucial role of obtaining ultrahigh-Q QBICs. Remarkably, we demonstrate that maintaining high Q-factors of QBICs hinges on perturbations within the minimum field region, while perturbations in the maximum field region lead to rapid Q-factor degradation. Experimental validation by fabricating PCSs with double holes and L-shaped holes corroborates these findings, showcasing QBIC Q-factors exceeding 30,000 and peaking at 36,394 when the structure perturbation aligns with the zero field region. Our results hold great promise for the design of high-performance photonic and optoelectronic devices, spanning applications such as lasing, sensing, and achieving strong coupling effects. By unlocking the full potential of SP BICs, we pave the way for boosting light-matter interaction and developing high-performance photonic and optoelectronic devices, such as lasing, sensing, and strong coupling.

5. Experimental Section

Simulation: The Q-factors and eigenfields of photonic crystal slabs were calculated by commercial software COMSOL Multiphysics based on the finite-element-method (FEM).

Fabrication: The samples were fabricated based on a silicon-on-insulator (SOI) wafer with a 220 nm top silicon layer and 2 µm buried layer, and the thickness of the Si substrate is ≈700 µm. First, a resist layer (ZEP520) is spin-coated on a clean SOI wafer. Then, the periodic patterns were defined in the resist layer with electron-beam-lithography. By using the resist as a mask layer, the patterns were then transferred to the silicon diaphragm by the inductively coupled plasma technique. Finally, the remaining resist was removed with the Nmethyl-2-pyrrolidone (NMP) liquor. Note that the samples were fabricated in Tianjin H-Chip Technology Group Corporation.

Optical Characterization: Scattering spectra were measured using a home-built microscope spectroscopy system. Cross-polarization measurement was applied to measure the high-Q resonant modes of the fabricated metasurfaces. A super-continuous laser (YSL SC-OEM) was used as a light source to provide a broad spectrum ranging from 430 to 2400 nm for full-frequency domain scanning. The laser beam first passed through the linear polarizer, the lens with a focal length of 200 mm, the beam splitter, and then was focused on the sample with an objective lens 10X. The reflected signal from the sample was split into two paths by the flipping mirror, one of which was used for imaging on CCD, and the other was collected by a spectrometer (YOKOGAWA AQ6370D) with a theoretical resolution of 0.02 nm. A pair of strictly orthogonal polarizers were placed at the front and the end of the light path, to introduce cross-polarization, which can efficiently eliminate the signal from direct reflection and thus enhance the signal-to-noise ratio.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

bound states in the continuum, dielectric metasurfaces, high-Q resonances, optical cavity

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